

Assignment 1

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1 Required Exercises

Exercise 1

The statement has the following tokens: `int a3 = a * 3 ;`

Total tokens: 7

Exercise 2

1、 $n+1$

2、 n

3、 1

4、 1

5、 1

6、 $\frac{n(n+1)}{2} + 1$

7、 2^n

Exercise 3

- 1、 The language of all strings over the alphabet $\{a, b\}$.
- 2、 The language of all strings over the alphabet $\{a, b\}$ that end with aaa , aab , aba or abb .
- 3、 The language of all strings over the alphabet $\{a, b\}$ where character b appears exactly 3 times.

Exercise 4

1、 $86-0755-[1-9][0-9]^7$

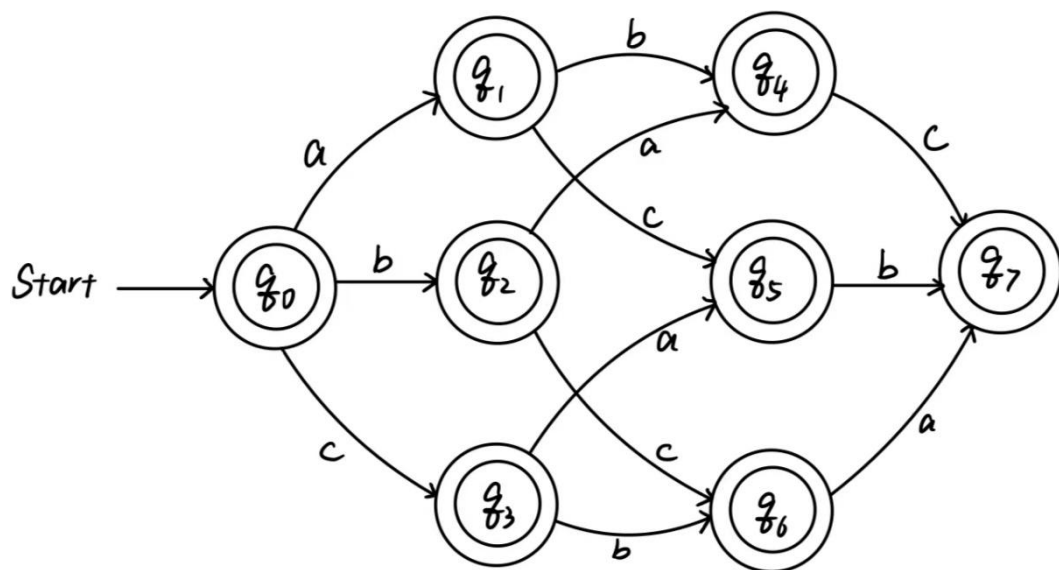
2、 $a(a|b)^*b$

3、 Let $consonant \rightarrow (b|c|d|f|g|h|i|k|l|m|n|p|q|r|s|t|v|w|x|y|z)^*$, the regular expression is $consonanta\ consonante\ consonanti\ consonanto\ consonantu\ consonant$.

2 Optional Exercises

Exercise 1

DFA:



The regular expression :

$(\epsilon|a)(\epsilon|b)(\epsilon|c)|$

$(\epsilon|a)(\epsilon|c)(\epsilon|b)|$

$(\epsilon|b)(\epsilon|a)(\epsilon|c)|$

$(\epsilon|b)(\epsilon|c)(\epsilon|a)|$

$(\epsilon|c)(\epsilon|a)(\epsilon|b)|$

$(\epsilon|c)(\epsilon|b)(\epsilon|a).$